J6 Series

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PRESSURE AND VACUUM SWITCHES
ADJUSTABLE DEADBAND MODELS

FEATURES

- Sensitive and Reliable; A Standard for Instrument Air Applications
- Gasketed, Die Cast Aluminum Enclosure with Epoxy Coating
- Adjustable Deadband Option
- SPDT Switch Output
- Adjustable Pressure Ranges:
  30 "Hg Vac to 6000 psi
  (-1 to 414 bar)
- Sealed, Isolated Metal Bellows Sensors

UNITED ELECTRIC CONTROLS

OVERVIEW

The UE J6 is a traditional pressure switch originally designed for instrument air applications in process plants. Its classic design features offer cost advantages.

The compact design offers a combination of sensitive On-Off operation and narrow or optional adjustable deadbands.

It is ideally suited for a wide range of industrial processes such as alarm/shutdown and low/high service pressures. OEMs also utilize the J6 in machinery and equipment for threshold protection.

FEATURES

- Sensitive and reliable; a standard for instrument air applications
- Gasketed, die cast aluminum enclosure with epoxy coating
- SPDT switch output
- Adjustable pressure ranges: 30 "Hg Vac to 6000 psi (-1 to 414 bar)
- NEMA 4X design
- Sealed, isolated metal bellows sensors

SPECIFICATIONS

STORAGE TEMPERATURE	-65° to 160°F (-54 to 71°C)
AMBIENT TEMPERATURE LIMITS	-40° to 160°F (-40 to 71°C)
SET POINT REPEATABILITY	Models 126-364, 680: ± 1% of adjustable range; models 610-614: ± 1 1/2% of adjustable range
SHOCK	Set point repeats after 15 G, 10 millisecond duration
VIBRATION	Set point repeats after 2.5 G, 5-500 Hz
ENCLOSURE	Die cast aluminum, epoxy powder coated, gasketed; captive cover screws
ENCLOSURE CLASSIFICATION	Designed to meet NEMA 4X requirements
SWITCH OUTPUT	One SPDT; switch may be wired “normally open” or “normally closed”; J6D has an adjustable deadband
ELECTRICAL RATING	15 A 125/250/480 VAC resistive
WEIGHT	Approx. 1 lb., 8 oz. (0.68 kg.)
ELECTRICAL CONNECTION	1/2” NPT female
PRESSURE CONNECTION	1/4” NPT female; models S126B-S160B: 1/2” NPT female

APPROVALS

UL listed
UL 508, file # E42272

CSA certified
CSA C22.2 no. 14, file # LR39690

CE Compliance to Low Voltage Directive (LVD)

PRESSURE MODEL CHART

Model	Stock #	Adjustable Set Point Range Low end of range on fall; High end of range on rise		Deadband		Over Range Pressure		Proof Pressure	
		psi	bar	psi	bar	psi	bar	psi	bar
316L welded stainless steel bellows with 1⁄2” NPT female pressure connection									
S126B	–	30 ″Hg Vac to 0 psi	-1 to 0	0.2 to 0.8 ″Hg	0,007 to 0,03	0	0	30 ″Hg Vac	-1,00
S134B	–	30 ″Hg Vac to 20 psi	-1 to 1,38	0.2 to 0.8 ″Hg	0,007 to 0,03	20	1,38	25	1,72
S136B	–	0 to 50″ wc	0 to 0,12	3 to 6 “wc	0,007 to 0,015	50 “wc	0,12	5	0,34
S142B	–	0 to 18	0 to 1,24	4 to 7 “wc	0,010 to 0,017	18	1,24	25	1,72
S148B	–	0 to 40	0 to 2,76	0.1 to 0.4	0,007 to 0,03	40	2,76	40	2,76
S152B	–	0 to 50	0 to 3,45	0.1 to 0.5	0,007 to 0,03	50	3,45	75	5,17
S156B	–	3 to 100	0,21 to 6,9	0.2 to 0.8	0,014 to 0,06	100	6,9	125	8,62
S160B	–	50 to 180	3,45 to 12,41	0.3 to 1	0,021 to 0,07	180	12,41	180	12,41
316L stainless steel bellows with 1⁄4” NPT female pressure connection (Model 680 not recommended for rapid or high cycling pressure changes)									
354	9596	0 to 50	0 to 3,45	1.5 to 2.5	0,10 to 0,17	50	3,45	75	5,17
356	9589	0 to 100	0 to 6,90	2 to 4	0,14 to 0,28	100	6,90	150	10,34
358	9646	0 to 200	0 to 13,79	3 to 5	0,21 to 0,34	200	13,79	250	17,24
360	–	0 to 250	0 to 17,24	3 to 5	0,21 to 0,34	250	17,24	330	22,75
362	9625	0 to 350	0 to to 24,13	2 to 8	0,14 to 0,55	350	24,13	430	29,65
364	9632	0 to 500	0 to 34,48	3 to 9	0,21 to 0,62	500	34,48	575	39,65
680	9582	100 to 1700	6,90 to 117	9 to 23	0,62 to 1,59	1700	117	2500	172
303 stainless steel piston and Buna N O-ring, 1⁄4” NPT female pressure connection (not recommended for gas service since drying of the O-ring can allow bleeding of the medium into the atmosphere)									
610	–	75 to 1000	5,17 to 69	30 to 150	2,07 to 10,34	1000	69	10,000	690
612	9716	125 to 3000	8,62 to 207	40 to 250	2,76 to 17,24	3000	207	10,000	690
614	–	500 to 6000	34,48 to 414	50 to 400	3,45 to 27,58	6000	414	10,000	690
Brass bellows with 1⁄4” NPT female nickel-plated brass pressure connection; Models 126 and 134 have zinc-plated spring in media									

Model	Stock #	Adjustable Set Point Range Low end of range on fall; High end of range on rise		Deadband		Over Range Pressure		Pressure Proof	
		wc	mbar	wc	bar	psi	bar	psi	bar
126	9543	30 ″Hg Vac to 0 psi	-1 to 0	0.2 to 0.8 ″Hg	0,007 to 0,03	0	0	30 ″Hg Vac	-1,00
134	9547	30 ″Hg Vac to 20 psi	-1 to 1,38	0.2 to 0.8 ″Hg	0,007 to 0,03	20	1,38	25	1,72
136	9501	0 to 50 ″wc	0 to 0,12	3 to 6 ″wc	0,007 to 0,015	50 ″wc	0,12	5	0,34
142	9508	0 to 18	0 to 1,24	4 to 7 ″wc	0,010 to 0,017	18	1,24	25	1,72
148	9522	0 to 40	0 to 2,76	0.1 to 0.4	0,007 to 0,03	40	2,76	40	2,76
152	9529	0 to 50	0,21 to 3,45	0.1 to 0.5	0,007 to 0,03	50	3,45	75	5,17
156	9536	3 to 100	0 to 6,89	0.2 to 0.8	0,014 to 0,06	100	6,9	125	8,62
160	–	50 to 180	3,45 to 12,41	0.3 to 1	0,021 to 0,07	180	12,41	180	12,41

Phosphor bronze bellows with 1⁄4″ NPT female nickel-plated brass pressure connection; Model 218 has 300 series stainless steel spring in media

218	96002	30 ″Hg Vac to 0 psi	-1 to 0	1 to 2 ″Hg	0,03 to 0,08	0	0	30	2,07
222	9561	0 to 20	0 to 1,38	0.5 to 1	0,03 to 0,07	20	1,38	30	2,07
224	9603	0 to 30	0 to 2,07	0.5 to 1	0,03 to 0,07	30	2,07	45	3,10
226	–	0 to 50	0 to 3,45	0.7 to 1.3	0,05 to 0,09	50	3,45	75	5,17
230	96016	0 to 100	0 to 6,90	1 to 2	0,07 to 0,14	100	6,90	110	7,58
258	9639	0 to 50	0 to 3,45	1.5 to 2.5	0,10 to 0,17	50	3,45	75	5,17
266	9610	0 to 100	0 to 6,90	2 to 5	0,14 to 0,34	100	6,90	150	10,34
270	9568	0 to 200	0 to 13,79	3 to 5	0,21 to 0,34	200	13,79	250	17,24
272	96030	0 to 250	0 to 17,24	3 to 5	0,21 to 0,34	250	17,24	330	22,75
274	9618	0 to 300	0 to 20,68	4 to 6	0,28 to 0,41	300	20,68	350	24,13

Type J6D

Standard adjustable deadband models; additional models are available with adjustable deadband by specifying option 1520.
Refer to options on page 6 for availability.

Brass bellows with 1⁄4″ NPT female nickel-plated brass pressure connection; Models 126 and 134 have zinc-plated steel spring in media

126	–	30 ″Hg Vac to 0 psi	-1 to 0	0.3 to 1.25 ″Hg	0,010 to 0,04	0	0	30 ″Hg Vac	-1,00
134	9541	30 ″Hg Vac to 20 psi	-1 to 1,38	0.3 to 1.25 ″Hg	0,010 to 0,04	20	1.38	25	1,72
142	9550	0 to 18	0 to 1,24	5 to 16 ″wc	0,01 to 0,04	18	1,24	25	1,72
148	–	0 to 40	0 to 2,76	0.1 to 0.8	0,007 to 0,06	40	2,76	40	2,76
156	9573	3 to 100	0, 21 to 6,89	0.5 to 2	0,03 to 0,14	100	6,89	125	8,62

HOW TO ORDER

BUILDING A PART NUMBER

Select a Type	Select a Model	Select an Option
Refer to the "Type" section below.	Refer to the "Model Charts".	Refer to the "Options" section.
Determine type number based on switch output, enclosure, adjustment and reference.	Determine model or stock number based on adjustable range, deadband and proof pressure.	Determine option number based on switch output, optional materials or other product enhancements.
Fill in the type portion of your part number with the corresponding number.	Fill in the model portion of your part number with the corresponding number.	Fill in the option portion of your part number with the corresponding number.
		Leave "option" portion blank if no options are needed. *FOR MULTIPLE OPTIONS:* Call United Electric Controls.

TYPE	DESCRIPTION
Pressure	**Type J6 - One** SPDT output; epoxy coated enclosure; internal adjustment with no reference dial
	Type J6D - Wide adjustable deadband; one SPDT output; epoxy coated enclosure; internal adjustment with no reference dial

SWITCH OPTIONS

0140	Gold flashed contacts, 1 A 125 VAC resistive (low energy circuits)
0500	Close deadband, 5 A 125/250 VAC resistive
1070	10 A 125 VDC resistive; deadband and minimum set point will increase
1520	Adjustable deadband, 15 A 125/250/277 VAC resistive. NOT AVAILABLE ON MODELS 258-274, 354-680, 610-614 (J6D INCLUDES OPTION)
1530	External manual reset, 15 A125/250/480 VAC resistive, latches on rising pressure only
2000	20 A 125/250 VAC resistive

SENSOR AND OTHER OPTIONS

M201	Factory set one switch; specify increasing or decreasing pressure and set point
M276	Range indicated on nameplate in bars/mbars
M278	Range indicated on nameplate in Kg/cm^2
M444	Paper ID tag
M446	Stainless steel ID tag & wire attachment
M540	Viton® construction; wetted parts include Viton® O-ring and standard connection material. AVAILABLE ON MODELS 610-614
M550	Oxygen service cleaning; internal construction may change

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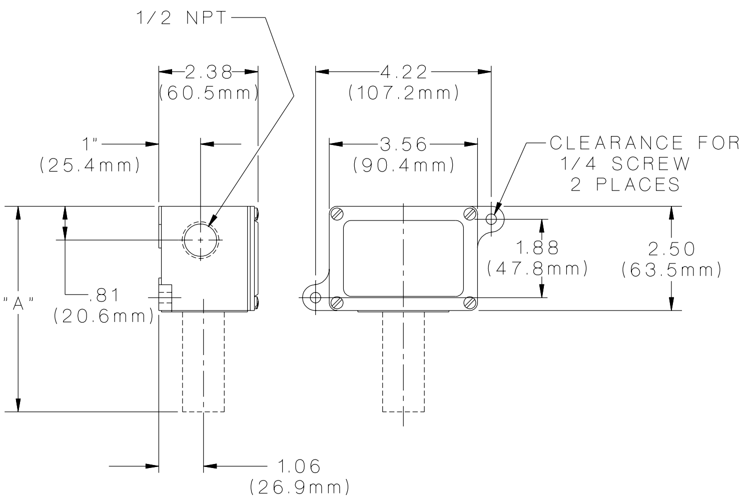
DIMENSIONAL DRAWINGS

J6 Series
General Purpose Service

Internal Set Point Adjustment

Types J6, J6D

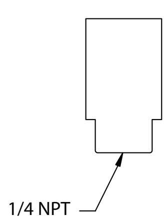


Dimension A			
Models	Inches	mm	NPT
126-160	5.06	128,5	1/4
S126B-S160B	5.40	137,2	1/2
218-230	4.31	109,5	1/4
258-274	4.75	120,7	1/4
354-364	4.75	120,7	1/4
610-614	5.69	144,5	1/4
680	4.5	114,3	1/4

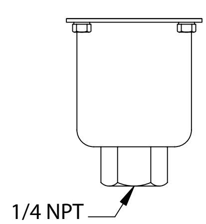
All dimensions stated in inches (millimeters)

Pressure Sensors

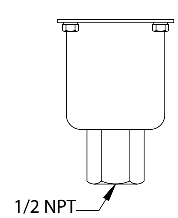
Models 218-230



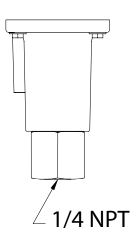
Models 126-160



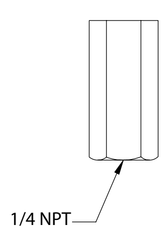
Models S126B-S160B



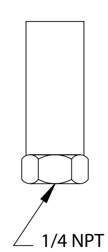
Models 610-614



Models 258-274



Models 354-364, 680



RECOMMENDED PRACTICES AND WARNINGS

United Electric Controls Company recommends careful consideration of the following factors when specifying and installing UE pressure and temperature units. Before installing a unit, the Installation and Maintenance instructions provided with unit must be read and understood.

- To avoid damaging unit, proof pressure and maximum temperature limits stated in literature and on nameplates must never be exceeded, even by surges in the system. Operation of the unit up to maximum temperature is acceptable on a limited basis (i.e., start-up, testing) but continuous operation must be restricted to the designated adjustable range. Excessive cycling at maximum temperature limits could reduce sensor life.
- A back-up unit is necessary for applications where damage to a primary unit could endanger life, limb or property. A high or low limit switch is necessary for applications where a dangerous runaway condition could result.
- The adjustable range must be selected so that incorrect, inadvertent or malicious setting at any range point cannot result in an unsafe system condition.
- Install unit where shock, vibration and ambient temperature fluctuations will not damage unit or affect operation. Orient unit so that moisture does not enter the enclosure via the electrical connection. When appropriate, this entry point should be sealed to prevent moisture entry.
- Unit must not be altered or modified after shipment. Consult UE if modification is necessary.
- Monitor operation to observe warning signs of possible damage to unit, such as drift in set point or faulty display. Check unit immediately.
- Preventative maintenance and periodic testing is necessary for critical applications where damage could endanger property or personnel.
- For all applications, a factory set unit should be tested before use.
- Electrical ratings stated in literature and on nameplate must not be exceeded. Overload on a switch can cause damage, even on the first cycle. Wire unit according to local and national electrical codes, using wire size recommended in installation sheet.
- Use only factory authorized replacement parts and procedures.
- Do not mount unit in ambient temp. exceeding published limits.

LIMITED WARRANTY OF REPAIR AND REPLACEMENT

Seller warrants that the product hereby purchased is, upon delivery, free from defects in material and workmanship and that any such product which is found to be defective in such workmanship or material will be repaired or replaced by Seller (F.O.B. UE Watertown); provided, however, that this warranty applies only to equipment found to be so defective within a period of 18 months from the date of manufacture by the Seller (36 months for the Spectra 12 and One Series products). Seller shall not be obligated under this warranty for alleged defects which examination discloses are due to tampering, misuse, neglect, improper storage, and in any case where products are disassembled by anyone other than authorized Seller's representatives.

EXCEPT FOR THE LIMITED WARRANTY OF REPAIR AND REPLACEMENT STATED ABOVE, SELLER DISCLAIMS ALL WARRANTIES WHATSOEVER WITH RESPECT TO THE PRODUCT, INCLUDING ALL IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PARTICULAR PURPOSE.

LIABILITY LIMITATION

SELLER'S LIABILITY TO BUYER FOR ANY LOSS OR CLAIM, INCLUDING LIABILITY INCURRED IN CONNECTION WITH (I) BREACH OF ANY WARRANTY WHATSOEVER EXPRESSED OR IMPLIED, (II) A BREACH OF CONTRACT, (III) A NEGLIGENT ACT OR ACTS (OR NEGLIGENT FAILURE TO ACT) COMMITTED BY SELLER, OR (IV) AN ACT FOR WHICH STRICT LIABILITY WILL BE IMPUTED TO SELLER, IS LIMITED TO THE LIMITED WARRANTY OF REPAIR AND REPLACEMENT STATED HEREIN. IN NO EVENT SHALL THE SELLER BE LIABLE FOR ANY SPECIAL, INDIRECT, CONSEQUENTIAL OR OTHER DAMAGES OF A LIKE GENERAL NATURE, INCLUDING, WITHOUT LIMITATION, LOSS OF PROFITS OR PRODUCTION, OR LOSS OR EXPENSES OF ANY NATURE INCURRED BY THE BUYER OR ANY THIRD PARTY.

UE specifications subject to change without notice.

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